stc89c52
The STC89C52 is a low-power, high-performance 8-bit 8051 Microcontroller manufactured by STC Microcontroller.

CPU

The STC89C52 features an enhanced 8-bit 8051-core central processing unit (CPU) that handles data chunks in 8-bit segments and operates at a maximum clock frequency of up to 35MHz to 48MHz.
cpu contains
-arithmetic logic units
-registers
-program counter
-stack pointer
-control unit
-cpu internal data path

REGISTERS
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they are small,fast,storage locations. it is used to hold data,addresses,intermediate results, and status information while a program is running.

examples;-

A(Accumulator)-mainly used for arithmetic, logical, and data operations.

B-used during multiplication and division operations.

R0–R7-: Eight 8-bit general-purpose registers used to temporarily store data. The 8051 architecture provides four register banks, with one bank selected at a time.

PSW-An 8-bit register that stores status flags such as Carry and Overflow and also selects the active register bank.

DATA POINTER(DPTR)-A 16-bit register used to store memory addresses.

PROGRAM COUNTER(PC)-A 16-bit register that stores the address of the next instruction to be executed.

STACK POINTER(SP)-An 8-bit register that keeps track of the current location of the stack.

PROGRAM COUNTER

The Program Counter (PC) is a 16-bit register in the STC89C52. It stores the address of the next instruction that the CPU needs to execute.

When the CPU starts running a program, the PC points to the location of the first instruction. After an instruction is fetched, the PC normally moves to the address of the next instruction.

STACK POINTER

The Stack Pointer (SP) is an 8-bit register in the STC89C52. It stores the address of the current top of the stack in internal RAM.
It is mainly used with instructions such as PUSH, POP, CALL, and RET.

MEMORY ORGANIZATION

The STC89C52 has separate areas for storing the program and data. The main memory areas are Program Memory-
Program memory stores the instructions of the program. The CPU reads instructions from this memory and executes them.

Internal RAM-
Internal RAM is used to store temporary data while the program is running. It contains areas for:

Register banks
Bit-addressable memory
General-purpose RAM
Stack

Special Function Registers (SFRs)-
SFRs are used to control and monitor different parts of the microcontroller. For example, they are used for I/O ports, timers, interrupts, and serial communication.

DATA MEMORY

Data memory is used to store and manage data during program execution. The STC89C52 has internal RAM that the CPU can use for temporary data storage.

I/O PORTS

The STC89C52 has four 8-bit I/O ports: P0, P1, P2, and P3. These ports allow the microcontroller to communicate with external devices.

Each port has 8 pins, so there are 32 I/O pins in total.

TIMERS/COUNTERS

The STC89C52 has three timers/counters: Timer 0, Timer 1, and Timer 2.A timer/counter is used to measure time or count events.

Timer: Counts internal clock pulses to measure time.
Counter: Counts external events or pulses.

INTERRUPT SYSTEMS

The interrupt system allows the STC89C52 to temporarily stop its current program and respond to an important event.

HOW INSTRUCTIONS ARE FETCHED AND EXECUTED

The STC89C52 CPU follows a basic Fetch–Decode–Execute process to run a program.

FETCH

The Program Counter (PC) contains the address of the next instruction. The CPU uses this address to read the instruction from program memory.

DECODE

The CPU identifies what the instruction means and determines what operation needs to be performed.

EXECUTE

The CPU performs the required operation using the ALU, registers, or memory.

UPDATE PC

The CPU performs the required operation using the ALU, registers, or memory.

** FEW IMPORTANT INSTRUCTION IN STC89C52**

MOV Moves data from one location to another
ADD Adds a value to the accumulator
SUBB Subtracts a value from the accumulator
INC Increases a value by 1
DEC Decreases a value by 1
MUL Multiplies A and B
DIV Divides A by B
SJMP Jumps to another location in the program
PUSH Stores data on the stack
POP Retrieves data from the stack